explain Preliminary Design Review in details and sub steps

Preliminary Design Review (PDR) is the gate where you check whether the chosen concept has been developed far enough—and in the right way—to justify full detailed design and prototyping. Below is a structured breakdown with practical sub steps.[1] [2]

1. Define PDR objectives and criteria

- Clarify the purpose of this PDR: confirm that system requirements are correctly allocated to subsystems, the architecture is sound, and major risks are understood with mitigation plans.[2] [1]
- Set entrance and exit criteria (e.g., required design maturity, analyses available, documents ready) and communicate them to all participants before the review.[3] [4]

2. Prepare design baseline and documentation

- Compile the preliminary design package: system architecture, block diagrams, major assemblies, key interfaces, performance budgets, and preliminary calculations or simulations.[5] [1]
- Include supporting plans and structures: product tree/WBS, updated technical specifications, verification plan, and engineering management plans (config-management, risk, quality).[1] [3]

3. Verify requirements allocation and traceability

- Show how each top-level requirement from the Requirements phase is allocated to subsystems/components, with traceability maintained in a matrix or similar tool.[2] [1]
- Check that the design can, in principle, meet each requirement (capacity, accuracy, interfaces, environmental limits, safety, reliability), and identify any requirements at risk. [6] [5]

4. Review system architecture and functional design

- Present functional descriptions and block diagrams that show major functions, data/energy flows, and control logic.[7] [5]
- Confirm that the architecture is complete, avoids unnecessary complexity, and respects constraints such as single-point-failure policies, redundancy, and scalability.[5] [2]

5. Evaluate key technical aspects and analyses

- Assess whether key engineering analyses for this maturity level have been done: structural, thermal, electrical, performance sizing, and any critical simulations.[1] [5]
- Review early results from reliability, safety, and hazard analyses, plus any environmental, thermal, or calibration considerations that will drive detailed design.[8] [5]

6. Check interfaces and compatibility

- Examine mechanical, electrical, software, and data interfaces between subsystems and to external systems (mounting points, connectors, communication protocols, envelopes).[8] [5]
- Verify that interface requirements (loads, mass, power, memory, timing, data rates, space claims) are defined, feasible, and consistent across all documents.[8] [5]

7. Assess producibility, materials, and make-or-buy

- Review major material choices, manufacturing processes, tolerancing approaches, and whether the design seems producible with available capabilities.[5] [8]
- Decide preliminary make-or-buy for key components and identify long-lead parts, special tooling, or facilities that need early action.[8] [5]

8. Review verification and test strategy

- Present the draft verification plan: which requirements will be shown by analysis, simulation, inspection, qualification tests, and acceptance tests.[1] [5]
- Check that the planned testability is built into the design (access for sensors, calibration features, built-in self-test, interfaces for diagnostics).[5] [8]

9. Analyze project risks, schedule, and resources

- Summarize key technical and programmatic risks identified so far, with likelihood/impact assessments and mitigation plans.[4] [3]
- Confirm that schedule, budget, and resource estimates for the next phases (detailed design, prototyping, testing) are realistic given design complexity and risk profile.[3] [6]

10. Conduct the review meeting

- Ensure all relevant stakeholders (engineering, manufacturing, quality, testing, project management, suppliers as needed) participate and receive the review package in advance. [4] [3]
- During the meeting, walk through the design package against the PDR objectives, capture Requests for Action (RFAs), findings, and observations in a structured way.[6] [4]

11. Decide outcome and actions

- Based on evidence, decide whether to: pass PDR and proceed to Detailed Design, pass with actions, require a re-review, or redirect/stop the project.[3] [2]
- Assign owners and due dates for all RFAs, required analyses, design changes, and documentation updates before the next milestone.[9] [4]

12. Document and baseline the preliminary design

- Record minutes, findings, decisions, and updated risk status, and store them together with the approved preliminary design package.[9] [4]
- Baseline the preliminary design (allocated baseline) so that future changes go through controlled change management and can be traced back to PDR decisions.[2] [1]

If you like, I can now turn these 12 sub steps into a compact yes/no checklist specifically tailored to mechanical systems (loads, tolerances, safety factors, DFMA, etc.) for your app.

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1. https://www.sciencedirect.com/topics/engineering/preliminary-design-review
2. https://acqnotes.com/acqnote/acquisitions/preliminary-design-review
3. https://www.dau.edu/acquipedia-article/preliminary-design-review-pdr
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